1. Permanent Magnet Moving Coil (PMMC) Instrument

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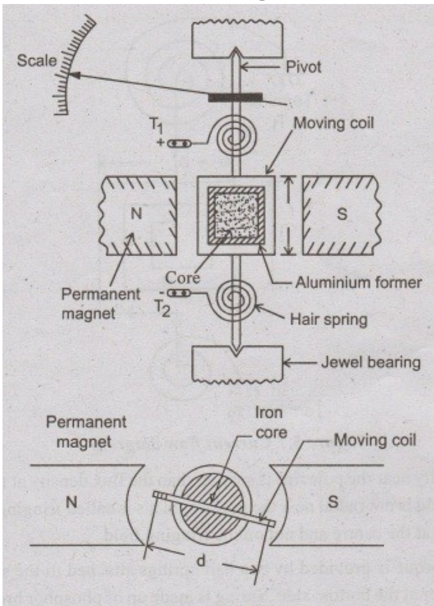

Figure 5.6 PMMC

Construction:

- **Permanent Magnet** → Provides a strong uniform magnetic field
- **Moving Coil** → Placed between magnet poles, wound on aluminum frame
- **Soft Iron Core** → Improves magnetic field strength
- **Spring (Control Spring)** → Provides controlling torque
- **Pointer & Scale** → Shows reading
- **Damping** → Eddy current damping (due to aluminum frame)

Principle:

- Works on **Lorentz force principle**:
  A current-carrying conductor placed in a magnetic field experiences a force.

Operation:

1. When **DC current flows** through the coil → magnetic field is produced.
2. Interaction with permanent magnetic field → **force acts on coil**.
3. Coil rotates → pointer moves on scale.
4. Spring provides opposing torque → system stabilizes.

Key Formula

Deflecting Torque (Td):

$$T_d = B \cdot I \cdot N \cdot A$$

Where:

- B = magnetic flux density
- I = current
- N = number of turns
- A = area of coil

Controlling Torque (Tc):

$$T_c = k \cdot \theta$$

Where:

- k = spring constant
- θ = deflection angle

At Equilibrium:

$$T_d = T_c$$

$$BINA = k\theta$$

$$\theta \propto I$$

Important Result:

✅ **Deflection is directly proportional to current**
→ **Uniform scale** (very important point for exams)

Advantages:

- High accuracy
- Uniform scale
- Low power consumption
- Good damping

Disadvantages:

- Works only with **DC (Direct Current)**
- Cannot measure AC directly

Applications:

- DC ammeters
- DC voltmeters
- Galvanometers

◆ 2. Extension of Range (Voltmeter, Ammeter, Ohmmeter)

➤ Ammeter Range Extension

- Use **Shunt resistor (parallel)**
- Formula:

$$R_s = \frac{I_m R_m}{I - I_m}$$

➤ Voltmeter Range Extension

- Use **Series resistor (Multiplier)**

$$R_s = \frac{V}{I_m} - R_m$$

➤ Ohmmeter Range

- Adjust internal battery + resistors
- Measures resistance based on current change

◆ 3. Sensitivity of Meter

➤ Definition

Sensitivity = resistance per volt

$$S = \frac{1}{I_m}$$

➤ Unit

Ohm/Volt

➤ Key Point

Higher sensitivity → more accurate instrument

4. Moving Iron Instruments (Attraction & Repulsion)

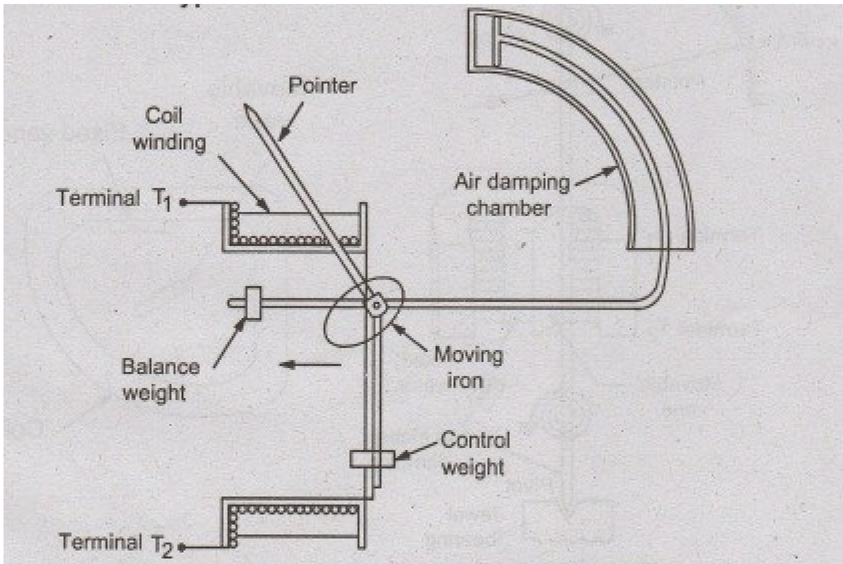

Figure 5.8 Attraction type moving iron instrument

1. Moving Iron Instruments (Repulsion & Attraction)

MI instruments basically measure **current or voltage** based on torque produced.

- **Torque (T) ∝ Current² (I^2)**

$$T \propto I^2$$

- **Deflection angle (θ) ∝ Torque**

$$\theta = k \cdot I^2$$

Where k is a constant depending on spring stiffness.

Moving Iron (MI) instruments are commonly used for measuring **AC and DC currents and voltages**. They work on the principle that a piece of iron gets magnetized when placed in a magnetic field and experiences a mechanical force. MI instruments are of **two types**: **Repulsion type** and **Attraction type**, each with slightly different operation. Here's a simple explanation:

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1. Repulsion Type Moving Iron Instruments

Construction & Principle:

- Contains **two iron pieces**, one fixed and one movable.
- When current flows through the coil, both iron pieces are magnetized with **like poles facing each other**.

- Like poles **repel** each other.

Function:

- The **movable iron piece rotates** due to magnetic repulsion.
- Rotation is opposed by a **spring**, which gives a **deflection proportional to current/voltage**.
- Used to measure **AC or DC**.

Characteristics:

- Better accuracy at **low currents**.
- Less friction because movable part is light.

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2. Attraction Type Moving Iron Instruments

Construction & Principle:

- Contains **one fixed coil and a movable iron vane/piece**.
- The coil is energized by the measured current.
- The iron is **attracted** towards the coil due to magnetization.

Function:

- The **movable iron moves toward the coil**.
- A **spring provides a counter torque**, so the deflection is proportional to current/voltage.
- Can also measure **AC or DC**.

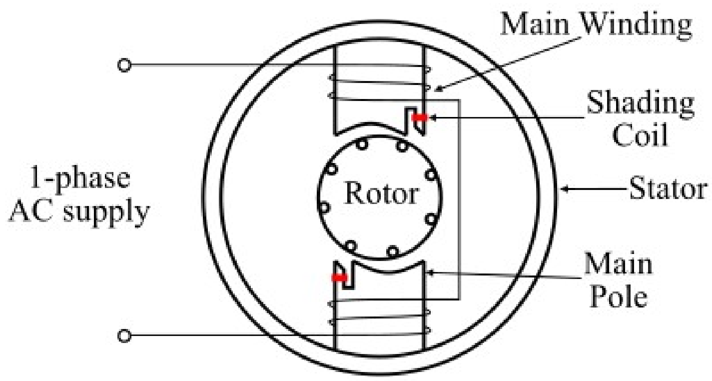
Characteristics:

- Simpler in construction.
- Generally used for **low-cost instruments**.

Key Difference Between Repulsion & Attraction MI Instruments

Feature	Repulsion Type	Attraction Type
Force	Repulsive (like poles repel)	Attractive (iron pulled to coil)
Accuracy	Higher at low currents	Lower accuracy
Construction	Two iron pieces	One iron piece
Common Use	Precision instruments	Low-cost instruments

5. Shaded Pole Induction Instruments & Energy Meter



2. Shaded Pole Induction Type Instruments

- **Torque on disc:**

$$T \propto I^2$$

- **Output deflection (θ):**

$$\theta = k \cdot I^2$$

Where I is AC current and k depends on disc inertia and spring constant.

Used for: AC ammeters, voltmeters, wattmeters.

1. Shaded Pole Induction Type Instruments

These instruments are mainly used to measure **AC current and voltage**. They are **AC-only** instruments.

Construction:

1. **Stator (Magnetic Core)**: Made of laminated soft iron to reduce eddy current losses.
2. **Coil**: A main coil wound on the stator to produce a magnetic field when AC flows.
3. **Shaded Pole**: A small copper ring (shading coil) around part of the pole. This **creates a delayed magnetic field** in that section.
4. **Moving Element (Rotor/Disc)**: Usually a lightweight aluminum disc mounted on a spindle.
5. **Control Spring**: Provides torque opposite to the rotation and brings the disc back to zero.
6. **Dial & Pointer**: Shows the measurement reading.

Operation:

- When AC passes through the coil, it produces an **alternating magnetic field** in the core.
- The **shading coil** delays the flux in that portion of the pole. This creates a **rotating magnetic field**.
- The **aluminum disc** experiences eddy currents due to this rotating flux.
- These **eddy currents produce torque**, causing the disc to rotate.

- The **rotation is proportional to AC current or voltage**. The pointer shows the reading on the dial.

Key Points:

- Works **only with AC**.
- Used in **AC ammeters, voltmeters, and wattmeters**.
- High frequency response is limited.

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2. Energy Meter (Induction Type)

3. Energy Meter (Induction Type)

- **Instantaneous Torque (T) ∝ V × I × cos φ**

$$T \propto VI \cos \phi$$

Where V = voltage across voltage coil, I = current in current coil, φ = phase angle between voltage and current.

- **Energy (kWh):**

$$\text{Energy} = \frac{\text{Number of disc revolutions}}{\text{Meter constant}}$$

Meter constant = revolutions per kWh.

Energy meters measure **electric energy consumption** (kWh) in homes, offices, or industries.

Construction:

1. **Stator**: Two coils
   - **Current coil (series)**: Connected in series with load, carries current.
   - **Voltage coil (shunt/parallel)**: Connected across supply voltage.
2. **Rotor (Aluminum Disc)**: Mounted on a spindle, free to rotate.
3. **Shading/Aluminum disc**: For generating torque via eddy currents.
4. **Brake Mechanism**: Usually an **aluminum disc eddy current brake** with a permanent magnet to control speed.
5. **Register/Counter**: Records the number of revolutions → corresponds to energy consumed.

Operation:

- When AC flows through the **current and voltage coils**, a **rotating magnetic field** is produced.
- **Eddy currents** are induced in the aluminum disc.
- These eddy currents generate **torque**, causing the disc to rotate.
- The **speed of rotation** is proportional to **power (V × I × cos φ)**.
- A **braking magnet** applies a retarding torque so that the disc rotates at a rate proportional to energy consumed, not instantaneous power.
- The **counter records energy in kWh**.

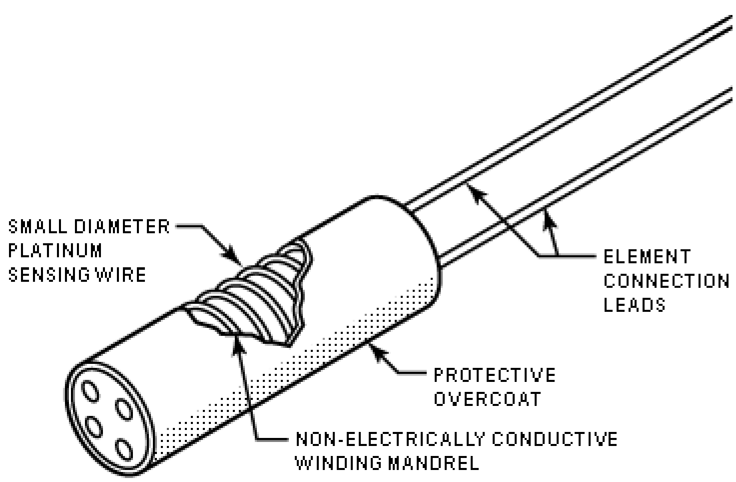
Key Points:

- Works **only with AC**.
- Measures **energy, not instantaneous power**.
- Used in **residential and industrial meters**.

Difference Between Shaded Pole Instrument & Energy Meter

Feature	Shaded Pole Instrument	Energy Meter (Induction Type)
Purpose	Measures AC current/voltage	Measures energy consumption (kWh)
Moving Part	Aluminum disc	Aluminum disc
Torque Generation	Eddy currents from AC & shaded pole	Eddy currents from current & voltage coils
Application	AC ammeters, voltmeters	Household/Industrial energy measurement
Control Mechanism	Spring	Braking magnet for speed control

◆ 6. Electrical Temperature Sensors



4. Electrical Temperature Sensors

a) RTD

$$R_t = R_0(1 + \alpha \cdot \Delta T)$$

Where:

- R_t = resistance at temperature T
- R_0 = resistance at 0°C
- α = temperature coefficient
- $\Delta T = T - 0^{\circ}C$

b) Thermocouple

$$V_{\text{out}} = k \cdot (T_{\text{measured}} - T_{\text{reference}})$$

Where k is Seebeck coefficient (mV/°C).

c) Thermistor

$$R = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

Where:

- R_0 = resistance at reference temperature T_0
- β = material constant
- T = absolute temperature (K)

1. RTD (Resistance Temperature Detector)

Construction:

- Made of **pure metal wire** (usually platinum, sometimes nickel or copper) wound on a ceramic or glass core.
- Housed in a **protective sheath** to prevent damage.
- Common types: **Pt100** (100 Ω at 0°C), Pt1000 (1000 Ω at 0°C).

Principle:

- Based on the **change in resistance of metal with temperature**.
- **Resistance increases with temperature** (positive temperature coefficient).

Operation:

- A constant current is passed through the RTD.
- The **voltage across the RTD** changes with resistance.
- This voltage is measured and converted into **temperature** using calibration tables or formulas.

Advantages:

- High accuracy.
- Stable over long periods.
- Good for **industrial applications**.

---

2. Thermocouple

Construction:

- Made of **two different metals** joined at one end (measuring/junction).
- Common types: **Type K (Nickel-Chromium / Nickel-Alumel), Type J (Iron / Constantan)**.

Principle:

- Based on the **Seebeck effect**: when two different metals are joined, a **voltage is generated that depends on temperature difference**.

Operation:

- Measuring junction is placed at the temperature to measure.
- The other end (reference junction) is kept at a known temperature.
- The **voltage generated** is proportional to the temperature difference.
- A meter or electronic circuit converts this voltage into **temperature reading**.

Advantages:

- Can measure **very high temperatures**.
- Fast response time.
- Simple and robust.

---

3. Thermistor

Construction:

- Made of **semiconductor material** (metal oxide ceramic).
- Shaped as **beads, discs, rods, or chips**.

Principle:

- Based on the **change of resistance with temperature**.
- Can have:
  - **NTC (Negative Temperature Coefficient)** → resistance **decreases** with temperature.
  - **PTC (Positive Temperature Coefficient)** → resistance **increases** with temperature.

Operation:

- Connected in a circuit (usually a voltage divider).
- The **change in resistance** alters the voltage across the thermistor.
- This voltage is measured and converted to **temperature**.

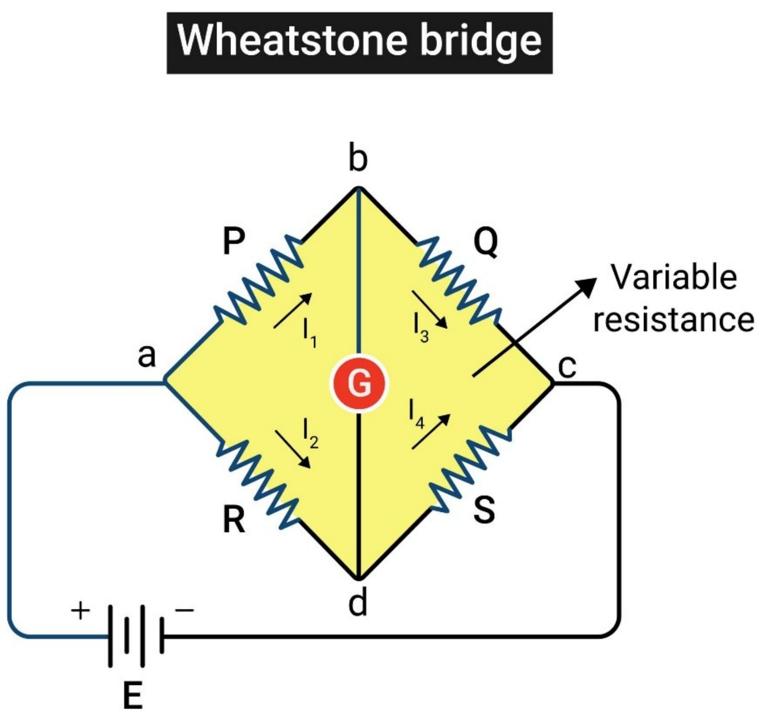
Advantages:

- Very sensitive to small temperature changes.
- Compact and inexpensive.
- Good for **precision temperature control**.

Comparison Table

Sensor Type	Principle	Temp. Range	Accuracy	Response	Application
RTD	Resistance changes w/ T	-200°C to 850°C	High	Medium	Industrial, Lab
Thermocouple	Seebeck effect (voltage)	-200°C to 2500°C	Medium	Fast	High-temp, furnaces
Thermistor	Resistance changes w/ T	-50°C to 150°C	Very High	Very Fast	Electronics, HVAC

◆ 7. Electrical Pressure Transducers



Strain Gauge Pressure Transducer

Construction:

- A **strain gauge** (resistive sensor) is attached to a **diaphragm**.
- The diaphragm deforms under applied pressure.
- Connected in a **Wheatstone bridge circuit** to detect resistance changes.

Principle:

- Based on **strain-resistance relationship**:
  - When the diaphragm is stressed by pressure, the strain gauge **stretches or compresses**.
  - Resistance of the strain gauge **changes proportionally**.

Operation:

1. Pressure applies force on the diaphragm.
2. Diaphragm bends → strain gauge deforms.
3. Resistance change in the gauge produces a **voltage change in the Wheatstone bridge**.
4. Output voltage is calibrated to show **pressure**.

Applications:

- Industrial process control, hydraulic systems, aerospace.

---

2. Potentiometric Pressure Transducer

Construction:

- Uses a **diaphragm** connected to a **slider of a potentiometer**.
- Potentiometer may be **linear or rotary type**.

Principle:

- Pressure deflects the diaphragm → moves the slider → changes **potentiometer resistance**.

Operation:

1. Pressure applied → diaphragm moves.
2. Slider moves along the potentiometer track.
3. Resistance changes → voltage output changes.
4. Voltage is calibrated to indicate **pressure**.

Applications:

- Low to medium pressure measurements, laboratory instrumentation.

---

3. Bourdon Tube Pressure Transducer

Construction:

- A **curved, hollow tube** (C or spiral shape).
- One end fixed, the other connected to a **mechanical linkage and pointer** or **electrical transducer**.

Principle:

- Pressure inside the tube causes **tube deformation (straightening or twisting)**.
- Mechanical movement is converted to **pointer deflection** or **electrical signal**.

Operation:

1. Fluid pressure enters the Bourdon tube.
2. Tube tends to straighten → moves a lever.
3. Lever moves pointer or electrical sensor (like potentiometer or LVDT).
4. Reading corresponds to pressure.

Applications:

- Analog pressure gauges, hydraulic systems, steam boilers.

---

4. LVDT (Linear Variable Differential Transformer) Pressure Transducer

Construction:

- Core connected to a **diaphragm**.
- LVDT consists of **primary coil and two secondary coils**.

Principle:

- Diaphragm deflection moves the **LVDT core**.
- This causes a **differential voltage** in the secondary coils → proportional to displacement → proportional to pressure.

Operation:

1. Pressure applies force on diaphragm.
2. Diaphragm moves → moves LVDT core.
3. LVDT generates an **AC voltage proportional to core displacement**.

4. Voltage output is calibrated to indicate pressure.

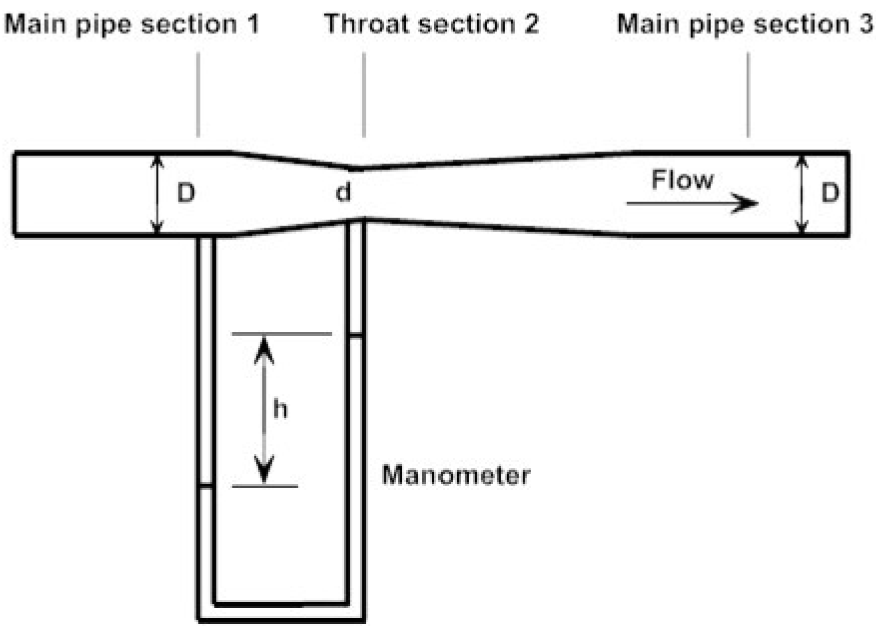
Applications:

- High accuracy pressure measurement, aerospace, industrial process control.

Comparison Table

Type	Principle	Accuracy	Applications
Strain Gauge	Resistance changes with strain	High	Industrial sensors, hydraulics
Potentiometric	Diaphragm moves potentiometer	Medium	Lab instruments, low-medium pressure
Bourdon Tube	Tube deformation	Medium	Analog gauges, boilers
LVDT-based	Displacement → voltage	Very High	Precise measurement, aerospace

8. Differential Flow Meters



➤ Differential Flow Meters

Purpose:

- Measure **flow rate of fluid** (liquid or gas) in a pipe.
- Work on the principle of **pressure difference** created by a flow restriction.

Principle:

- When fluid flows through a constriction (like orifice or venturi), **velocity increases and pressure decreases**.
- According to **Bernoulli's principle**, the **pressure difference** is proportional to **flow rate**.

---

1. Orifice Flow Meter

Construction:

- A **thin plate with a sharp-edged hole (orifice)** placed inside the pipe.
- Connected to **differential pressure measuring device** (manometer, transmitter).

Operation:

1. Fluid flows toward the orifice → **velocity increases**, **pressure drops** at the orifice.
2. Differential pressure is measured **before and at the orifice**.
3. Flow rate is calculated using Bernoulli's equation and **flow coefficient**.

Characteristics:

- Simple and low cost.
- Creates **high pressure loss**.
- Suitable for **low to medium accuracy requirements**.

---

2. Venturi Tube Flow Meter

Construction:

- A **gradually converging section → throat → gradually diverging section** in a pipe.
- Differential pressure measured **between inlet and throat**.

Operation:

1. Fluid enters **converging section → velocity increases**, **pressure decreases**.
2. At **throat**, maximum velocity → minimum pressure.
3. Pressure difference between inlet and throat is measured.
4. Flow rate calculated using Bernoulli's equation.

Characteristics:

- Smooth converging/diverging reduces **energy loss**.
- More **accurate** than orifice meters.
- More expensive and bulky.

Comparison Table

Feature	Orifice Meter	Venturi Tube Meter
Construction	Thin plate with hole	Converging-diverging tube
Pressure Loss	High	Low
Accuracy	Medium	High
Cost	Low	High
Applications	General industrial flow	High accuracy flow measurement, large pipelines

➤ **Flow Formula**

$$Q \propto \sqrt{P_1 - P_2}$$